INCH-POUND

MIL-PRF-39016/19G
AMENDMENT 7
6 October 2003
SUPERSEDING
AMENDMENT 6
7 September 1998

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT,
LOW LEVEL TO 1.0 AMPERE WITH INTERNAL DIODES FOR COIL
TRANSIENT SUPPRESSION AND POLARITY REVERSAL PROTECTION,
TERMINALS 0.100-INCH GRID PATTERN

This amendment forms a part of MIL-PRF-39016/19G, dated 20 July 1988, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

FIGURE 1: Delete ".370 SQ MAX" and substitute ".375 SQ MAX".

PAGE 2

FIGURE 2: Delete ".370 SQ MAX" and substitute ".375 SQ MAX".

FIGURE 2, mounting pad: Delete ".170 ±005 and substitute ".140 minimum, .175 maximum".

PAGE 3

REQUIREMENTS, contact data, high level (relay case grounded), resistive, delete and substitute:

Resistive:
1.0 ampere at 28 V dc.
250 milliamperes at 115 V ac 60 and 400 Hz case not grounded.
100 milliamperes at 115 V ac 60 and 400 Hz case grounded.

REQUIREMENTS, overload (high level only), delete and substitute:

Overload (high level only): Two times rated current. Not applicable to ac load ratings.

PAGE 4

ELECTRICAL DATA, dielectric withstanding voltage, delete and substitute:

	Sea level V rms (60 Hz)	Post intermediate current life test Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame or enclosure, and all contacts in the energized and deenergized positions: ----------	500	500	
Between case, frame or enclosure, and coil: ----------	500	500	125
Between all contacts and coil: ----------	500	500	All terminals
Between open contacts in the energized and deenergized positions: --	500	375	to case
Between contact poles in the energized and deenergized positions: ---	500	500	
Between coils of dual coil relays	NA	NA	

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ELECTRICAL DATA, delete footnote 1/ and associated references to footnote 1/in their entirety.

ELECTRICAL DATA, diode characteristics, delete and substitute:

"Coil transient suppression: Applicable.

Diode block integrity (perform this test after coil transient suppression test in all inspection tables of MIL-PRF-39016): With applicable voltage applied to the relay coil circuit in the reverse direction, monitor leakage current with dc microammeter, oscilloscope, or qualifying activity approved test equipment. Leakage current shall not exceed the specified value.

Block integrity maximum leakage current: ljiA at 50 V dc.

Maximum negative transient: 1.0 volt.

Breakdown voltage: 100 V dc minimum at 10 microamperes (MA). (This test may be performed in-process or as final assembly).

Semiconductor in-process screening: Applicable, visual inspection of semiconductors shall be in accordance with MIL-STD-750, methods 2073 or 2074."

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TABLE I, add the following new footnote 7/:

"7/ Delete "Coil resistance" and substitute "Coil current" test in all inspection tables of MIL-PRF-39016." TABLE I, add footnote 7/ to the coil circuit current column.

PAGE 7

* Group A testing for relay, delete: "Before installation of pad, perform subgroup 2 of group A tests. After installation of pad, perform subgroups 3 and 4 of group A tests." and substitute "Perform seal test immediately preceding the A2 electrical tests. Relay leads shall be formed and the mounting pad removed before the seal test. After the seal test, the mounting pad shall be rigidly attached to the relay and the remaining group A tests performed (The seal test is not performed with group A4.)."

The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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